

Learning Discriminative Representations to Interpret Image Recognition Models

Thèse de Doctorat

Felipe Torres Figueroa

École Centrale de Marseille

QARMA - Laboratoire d'Informatique et de Systèmes (LIS)

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President: Frédéric Jurie

Reviewers: Frédéric Jurie
Giorgos Tolas

Examiners: Frédéric Precioso
Diane Larlus

Supervisors: Stéphane Ayache
Ronan Sicre
Yannis Avrithis

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- Introduction
- 1 Opti-CAM: Optimizing saliency maps for interpretability
- 2 CA-Stream: Attention-based pooling for interpretable image recognition
- 3 A learning paradigm for interpretable gradients
- Closing Remarks

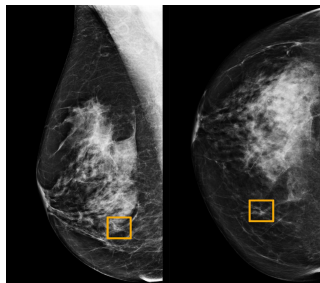
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Motivation

Introductory Examples

Consider these situations:



Motivation

Introductory Examples

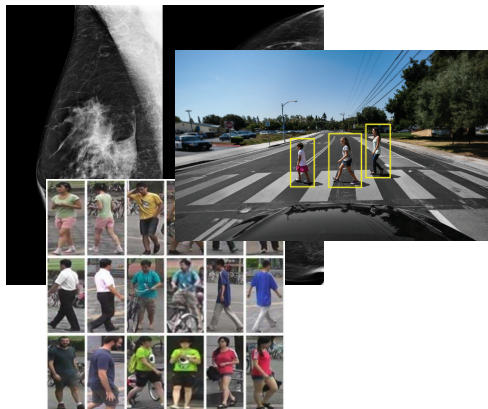
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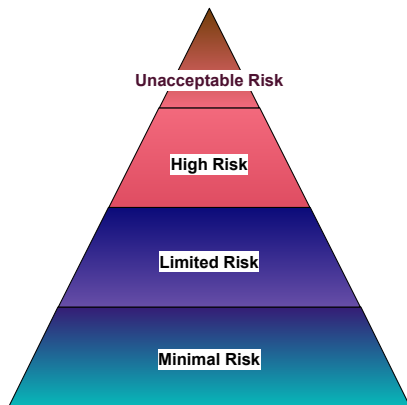
Introductory Examples

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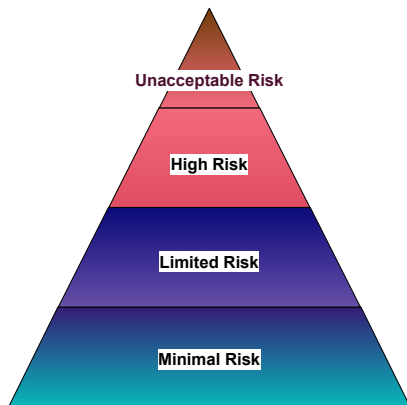
AI accepted in society?



- How do we **know how** a system works?
- How do we **know how** safe a system is?
- If a system fails, **who** is accountable?

Motivation

AI accepted in society?



- How do we **know how** a system works?
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We must **explain** the behaviour of these models and their predictions.

Image Recognition Models

An ongoing revolution since 2012

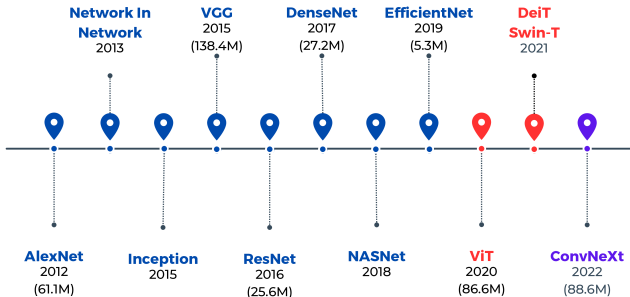
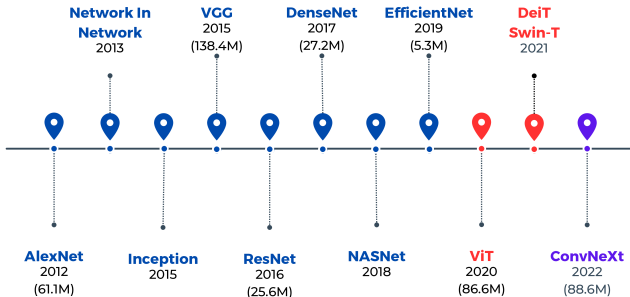


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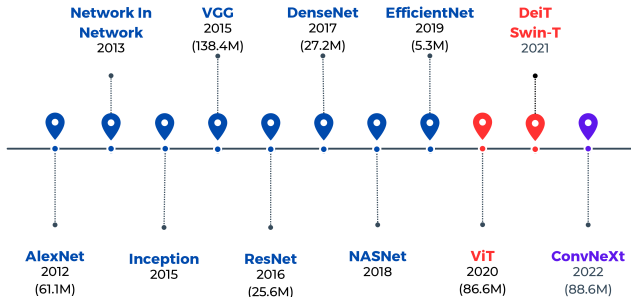


Contributing factors:

- Widespread of Internet.
- Big Dataset collection.
- Popularization of the GPU.

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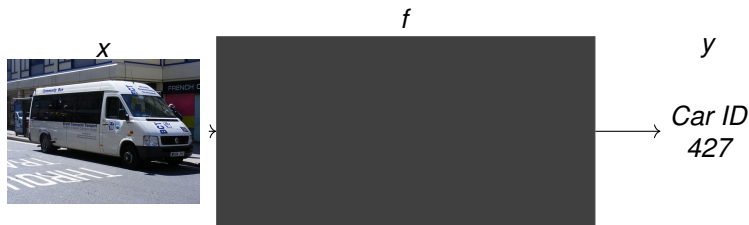
Two paradigms:

Convolutional Neural Networks (CNNs).

Transformers.

Motivation

We are interested in understanding models, behaving like a black box:



We want to *know why* $f(x) \rightarrow y$

Interpretability

A paradigm on interpretable studies

Interpretability approaches have been categorized by two studies [1, 2]:

1. **Passive or Active?**

To add or not modifications to the model.

Interpretable Studies Landscape

D1: Active or Passive



Interpretability

A paradigm on interpretable studies

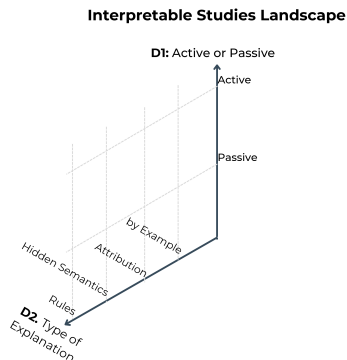
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Is it an example? An Attribution? Hidden Semantics? Or a Rule?



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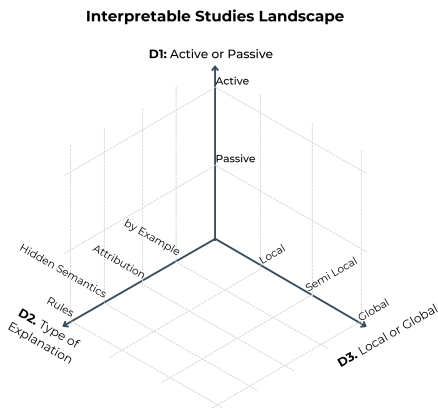
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3. **Local or Global?**

Is it a brief explanation? Does it cover more data/ of the model? Does it explain the model/data entirely?

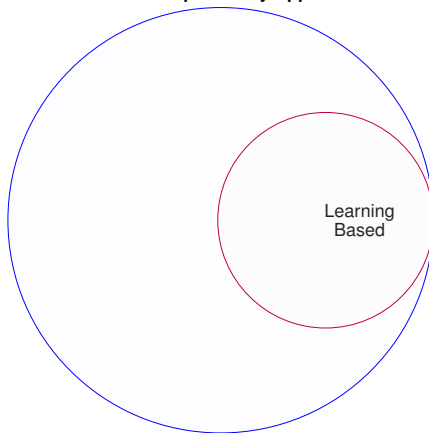


¹ Lipton defines Interpretability Similarly.
Transparency or Post-hoc Interpretations

Interpretability

Post-hoc Interpretability

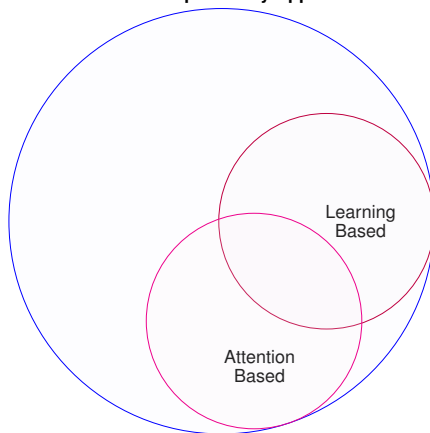
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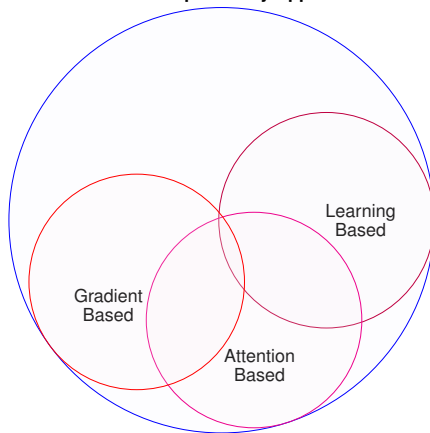
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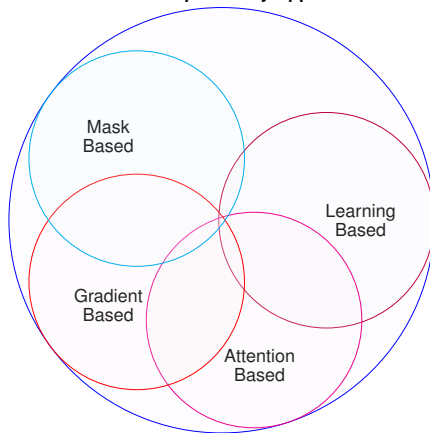
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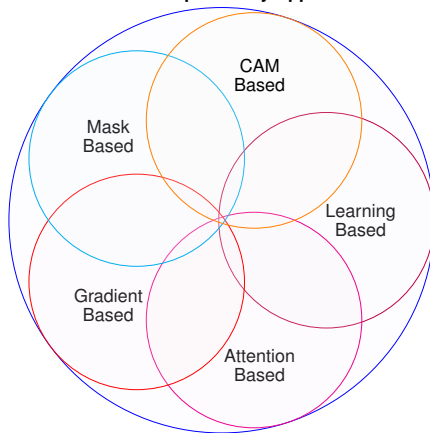
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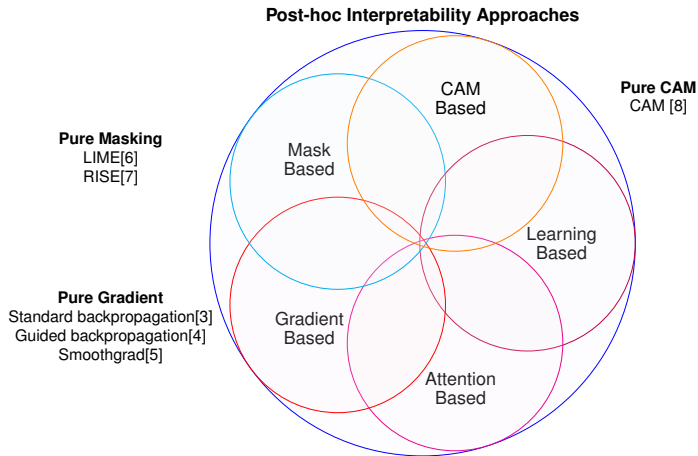
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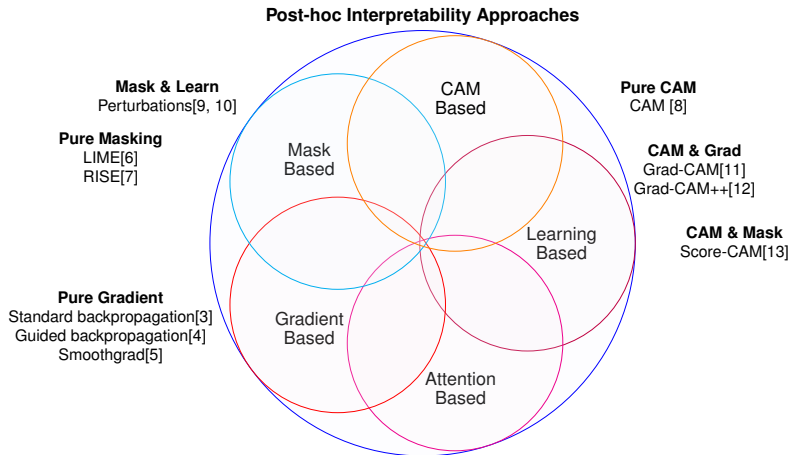
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Contributions

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We demonstrate improvements in these approaches

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- Optimized explanations.

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- Enhanced training for interpretability.

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Preliminaries

Premise

Is it possible to optimize a saliency map probability with the highest probability of prediction?



$$p(x)_{774} = 0.745$$



$$p(x)_{774}^{\text{CAM}} = 0.728$$

Preliminaries

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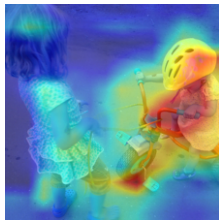
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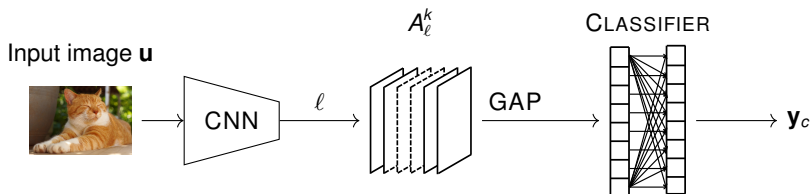
$$p(x)_{774}^{CAM} = 0.728$$



$$p(x)_{774}^{Optimal} = 0.853$$

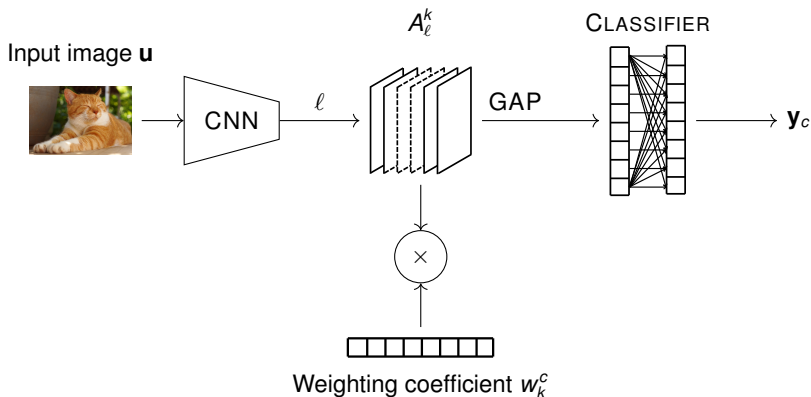
Background

CAM Methods



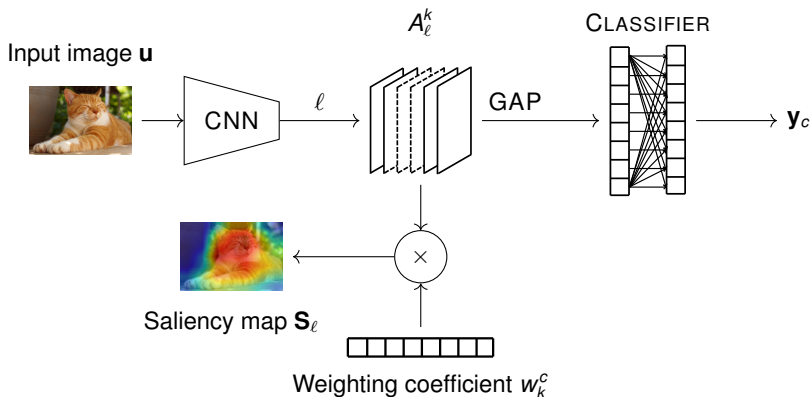
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CAM Methods



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CAM Formalized

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CAM[8]:

w_k^c comes from the weights
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[11, 12]:

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w_k^c is computed from the contributions of individual feature maps masking the input image.

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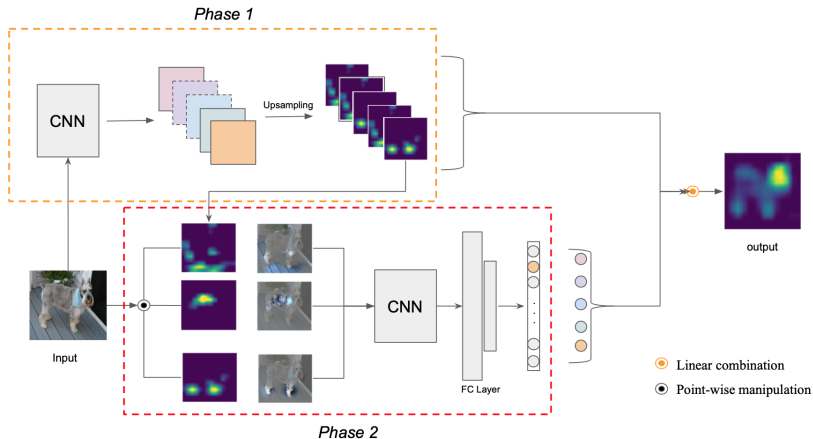
Score-CAM[13]:

w_k^c is computed from the contributions of individual feature maps masking the input image.

The approach found in Score-CAM is interesting. Diving deeper:

Background

Score-CAM



Original from [13]

Background

Score-CAM Formalized

Following equation (1):

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Score-CAM Formalized

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- $h := \text{ReLU}$
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Background

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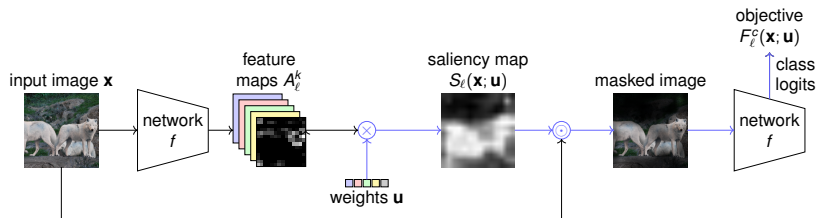
- $h := \text{ReLU}$
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With feature map contribution $\mathbf{u}_k^c$:

$$u_k^c := f(\mathbf{x} \odot n(\text{up}(A_\ell^k)))_c - f(\mathbf{x})_c \quad (2)$$

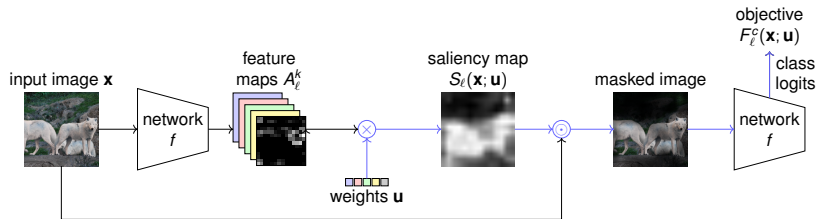
Opti-CAM

Approach



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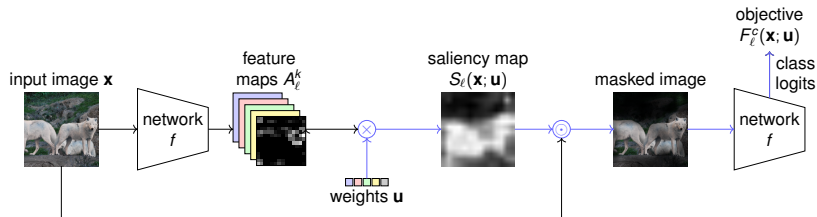


Consider S_ℓ , a function of $\mathbf{u}$ and $\mathbf{x}$:

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (3)$$

Opti-CAM

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Consider S_ℓ , a function of $\mathbf{u}$ and $\mathbf{x}$:

$$S_\ell(\mathbf{x}; \mathbf{u}) := \sum_k \text{softmax}(\mathbf{u})_k A_\ell^k \quad (3)$$

$\mathbf{u}^*$ is the vector that maximizes class c probability, when masking the the input with S_ℓ .

$$\mathbf{u}^* := \underset{\mathbf{u}}{\text{argmax}} F_\ell^c(\mathbf{x}; \mathbf{u}) \quad (4)$$

And the objective $F_\ell^c(\mathbf{x}; \mathbf{u}) := f(\mathbf{x} \odot n(\text{up}(S_\ell(\mathbf{x}; \mathbf{u}))))$

Evaluating Interpretability

Three paradigms

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- **Interpretable Image Recognition[12]**

Average Drop (AD), Average Increase (AI), Average Gain (AG).

Evaluating Interpretability

Three paradigms

- **Interpretable Image Recognition[12]**
Average Drop (AD), Average Increase (AI), Average Gain (AG).
- **Causality Analysis[7]**
Insertion & Deletion.
- **Weakly Supervised Localization Approach**

Evaluating Interpretability

Three paradigms

- **Interpretable Image Recognition[12]**

Average Drop (AD), Average Increase (AI), Average Gain (AG).

- **Causality Analysis[7]**

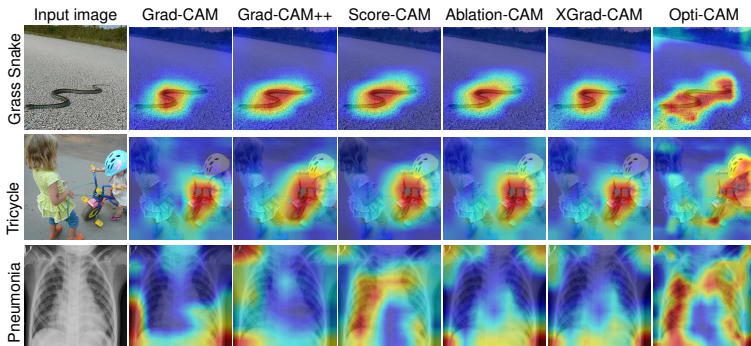
Insertion & Deletion.

- **Weakly Supervised Localization Approach**

Official Metric (OM), Localization Error (LE), Pixel-wise F_1 score (F1) Box Accuracy (BA), Standard Pointing Game (SP), Energy Pointing Game (EP).

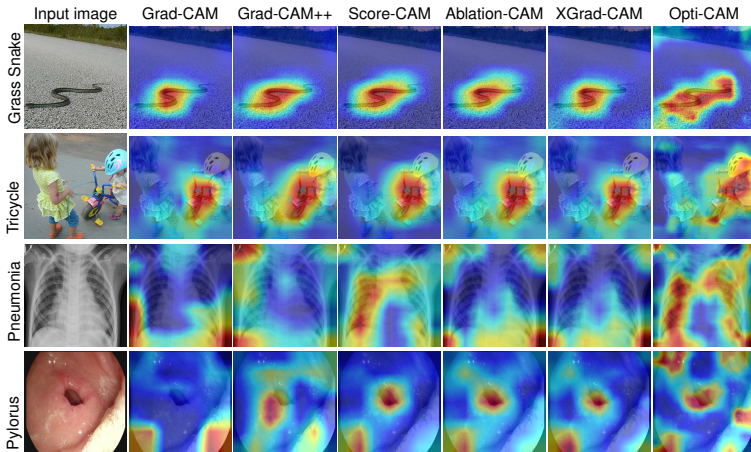
Results

Qualitative Evaluation



Results

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Results

Quantitative Evaluation

Classification Experiments:

METHOD	RESNET50						VGG16					
	AD ↓	AG ↑	AI ↑	I ↑	D ↓	T(s)	AD ↓	AG ↑	AI ↑	I ↑	D ↓	T(s)
Fake-CAM	0.8	1.6	46.0	50.7	28.1	0.00	0.5	0.6	42.6	46.1	26.9	0.00
Grad-CAM	12.2	17.6	44.4	66.3	14.7	0.03	14.2	14.7	40.6	64.1	11.6	0.02
Grad-CAM++	12.9	16.0	42.1	66.0	14.7	0.03	17.1	10.2	33.4	62.9	12.2	0.02
Score-CAM	8.6	26.6	56.7	65.7	16.3	15.22	13.5	15.6	41.7	62.5	12.1	3.11
Ablation-CAM	12.5	16.4	42.8	65.9	14.6	18.26	15.5	12.6	36.9	63.8	11.4	2.98
XGrad-CAM	12.2	17.6	44.4	66.3	14.7	0.03	13.8	14.8	41.2	64.1	11.7	0.02
Layer-CAM	15.6	15.0	38.8	67.0	14.2	0.08	48.9	3.1	13.5	58.3	6.4	0.07
ExPerturbation	38.1	9.5	22.5	70.7	15.0	152.96	43.0	7.1	20.5	61.1	15.0	83.20
Opti-CAM	1.5	68.8	92.8	62.0	19.7	4.15	1.3	71.2	92.7	59.2	11.0	3.94

Results

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The optimization objective is well aligned with AD, AI, AG.
Insertion and Deletion are better suited for sparse saliency maps.

Results

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Layer-CAM	73.1	66.0	50.1	55.5	70.0	33.0	1.29		70.5	61.5	28.0	54.7	65.0	32.4	1.45	
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Localization is hard. It acts more aligned to human interpretations than model ones.

Conclusions

- Classifier specific saliency maps. The probability of prediction of a class is the highest.
- No regularization needed unlike [9, 10]. Optimization only over feature dimensions.
- Localization is difficult for Opti-CAM. It is mostly human based.
- Average Gain complements the evaluation procedure.
- Published at Computer Vision and Image Understanding (CVIU) 2024.

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Premise

Is it possible to incorporate transformers ideas onto CNNs, enhancing the predictive power of CAM explanations?



$$p = 0.7623$$



$$p = 0.7145$$

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$$p = 0.7623$$



$$p = 0.7145$$



$$p = 0.7478$$

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Self Attention

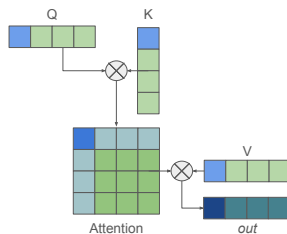
Self-attention is the building block of Transformers:

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Self Attention

Self-attention is the building block of Transformers:

$$\text{Self Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (5)$$

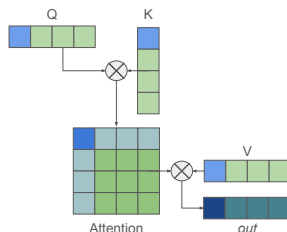


Background

Self Attention

Self-attention is the building block of Transformers:

$$\text{Self Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_e}} \right) V \quad (5)$$



An abstract class representation ([CLS]) updated with information from the embedding. [14]

Background

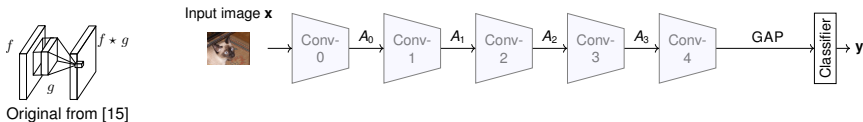
Convolutions and Self Attention

Different building blocks rely on different pooling mechanisms:

Background

Convolutions and Self Attention

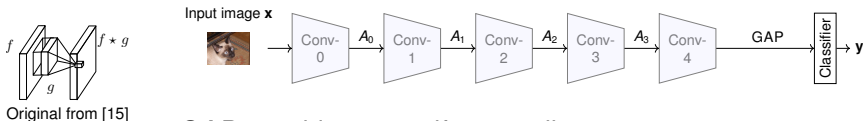
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Background

Convolutions and Self Attention

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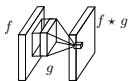


GAP considers an uniform pooling.

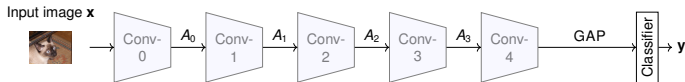
Background

Convolutions and Self Attention

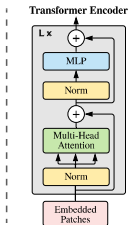
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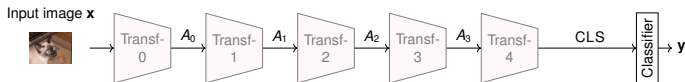
Original from [15]



GAP considers an uniform pooling.



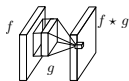
Original from [16]



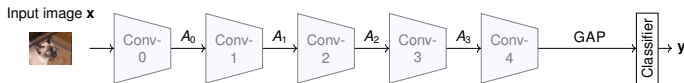
Background

Convolutions and Self Attention

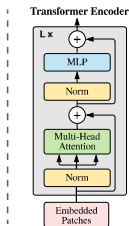
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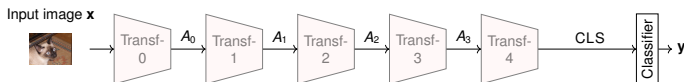
Original from [15]



GAP considers an uniform pooling.



Original from [16]



CLS is weighted with attention.

Motivation

Including CLS in CNNs

Cross Attention: Assuming an arbitrary embedding $\mathbf{q}$:

$$\text{CA}_\ell(\mathbf{q}_\ell, A_\ell) := A^\top \mathbf{q} = A_\ell^\top h_\ell(A_\ell \mathbf{q}_\ell), \in \mathbb{R}_\ell^d \quad (6)$$

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For a feature map A_ℓ , a pooled representation can be expressed via attention vector $\mathbf{a} = h_\ell(A_\ell \mathbf{q}_\ell)$:

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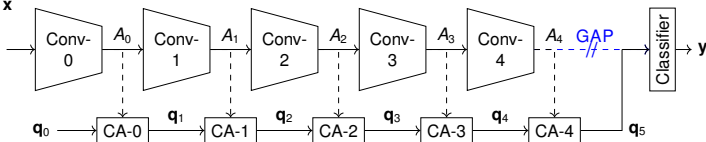
$$\text{CA}_\ell(\mathbf{q}_\ell, A_\ell) := A_\ell^\top \mathbf{a} \quad (7)$$

A pooling similar to GAP, masked in a fashion similar to CAM.

Cross Attention Stream

Given a model f we consider locations where critical operations take place:

Input image $\mathbf{x}$

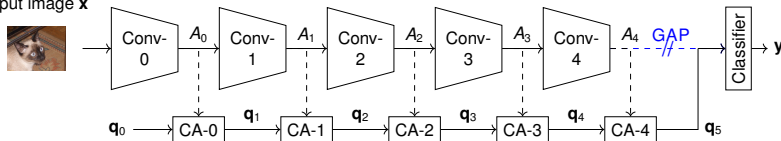


$$f = g \circ \text{GAP} \circ f_L \circ \dots \circ f_0 \quad (8)$$

Cross Attention Stream

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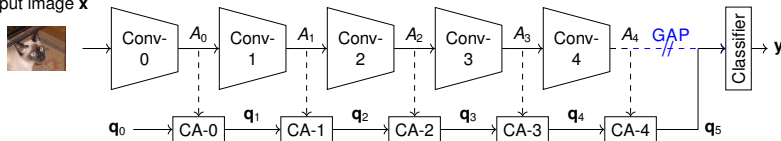
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A [CLS] token embedding is initialized as $\mathbf{q}_0 \in \mathbb{R}^{d_0}$.

Cross Attention Stream

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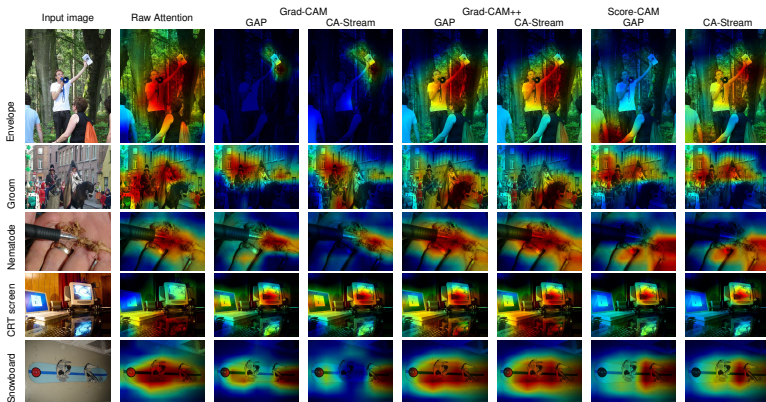
$$f = g \circ \text{GAP} \circ f_L \circ \dots \circ f_0 \quad (8)$$

A [CLS] token embedding is initialized as $\mathbf{q}_0 \in \mathbb{R}^{d_0}$. Updated with a sequence of embeddings $\mathbf{q}_\ell \in \mathbb{R}^{d_\ell}$, interacting along a stream with features on stage ℓ :

$$\mathbf{q}_{\ell+1} = W_\ell \cdot \text{CA}_\ell(\mathbf{q}_\ell, A_\ell) \quad (9)$$

Results

Qualitative Experiments



Results

Quantitative Experiments

Recognition results

ACCURACY AND PARAMETERS					
NETWORK	POOL	GFLOPS	#PARAM	PARAM%	ACC↑
RESNET-18	GAP	3.648	11.69M	3.71	67.28
	CA	3.652	12.13M		67.54
RESNET-50	GAP	8.268	25.56M	27.27	74.55
	CA	8.288	32.53M		74.70
CONVNEXT-S	GAP	17.395	50.22M	1.95	83.26
	CA	17.400	51.20M		83.14
CONVNEXT-B	GAP	30.747	88.59M	1.96	83.72
	CA	30.753	90.33M		83.51

CA Maintains recognition properties.

Results

Quantitative Experiments

NETWORK	METHOD	POOL	AD↓	AG↑	AI↑	I↑	D↓
RESNET-50	Grad-CAM	GAP	13.04	17.56	44.47	72.57	13.24
		CA	12.54	22.67	48.56	75.53	13.50
	Grad-CAM++	GAP	13.79	15.87	42.08	72.32	13.33
		CA	13.99	19.29	44.60	75.21	13.78
	Score-CAM	GAP	8.83	17.97	48.46	71.99	14.31
		CA	7.09	23.65	54.20	74.91	14.68
CONVNEXT-B	Grad-CAM	GAP	33.72	2.43	15.25	52.85	29.57
		CA	19.45	13.96	32.89	86.38	45.29
	Grad-CAM++	GAP	34.01	2.37	15.60	52.83	29.17
		CA	36.69	8.00	21.95	85.39	53.42
	Score-CAM	GAP	43.55	2.23	15.67	50.96	39.49
		CA	23.51	11.04	27.35	83.41	60.53

Results

Quantitative Experiments

NETWORK	METHOD	POOL	AD↓	AG↑	AI↑	I↑	D↓
RESNET-50	Grad-CAM	GAP	13.04	17.56	44.47	72.57	13.24
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	Grad-CAM++	GAP	34.01	2.37	15.60	52.83	29.17
		CA	36.69	8.00	21.95	85.39	53.42
	Score-CAM	GAP	43.55	2.23	15.67	50.96	39.49
		CA	23.51	11.04	27.35	83.41	60.53

CA obtains stronger representations for interpretability.
Deletion poses issues due OOD data.

Conclusions

- CAM behaves like Attention. Attention on final layers ought to behave like CAM.
- Pooling matters. Smarter pooling yields improved interpretations, maintaining recognition performance.
- Deletion requires a revisit. Zeros on images still provide a signal.
- Published in the XAI Workshop at CVPR 2024.

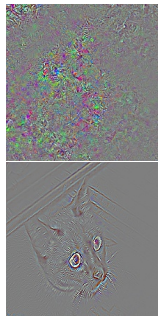
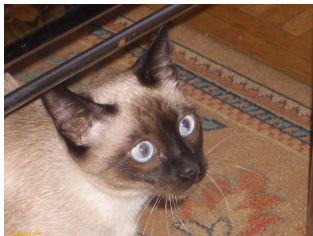
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- Introduction
- 1 Opti-CAM: Optimizing saliency maps for interpretability
- 2 CA-Stream: Attention-based pooling for interpretable image recognition
- 3 A learning paradigm for interpretable gradients**
- Closing Remarks

Preliminaries

Premise

Can smoothing gradients during training, improve interpretability properties?



Background

Leveraging Backpropagation

The response (y) of a model (f) parameters (ϕ), after forwarding an input x can be visualized on the input space, via backpropagation:

$$\frac{\partial x}{\partial y} = \frac{\partial x}{\partial F_0} \frac{\partial F_0}{\partial F_1} \cdot \dots \cdot \frac{\partial F_{-1}}{\partial y} \quad (10)$$

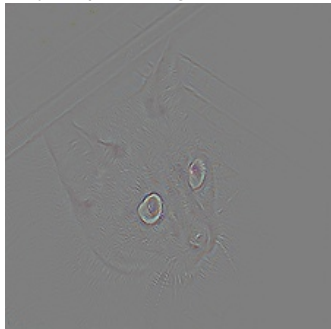
Background

Soft Gradients

Soft gradients can be achieved by two approaches:

Guided Backpropagation [4]

Achieved by constraining backprop to only let positive gradients flow:



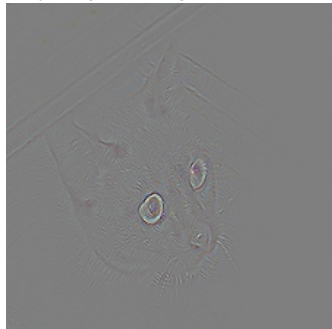
Background

Soft Gradients

Soft gradients can be achieved by two approaches:

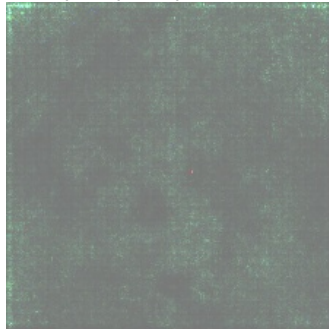
Guided Backpropagation [4]

Achieved by constraining backprop to only let positive gradients flow:



Smoothgrad [5]

Achieved by iteratively adding noise to the input space prior to forward pass:



Interpretable Gradients

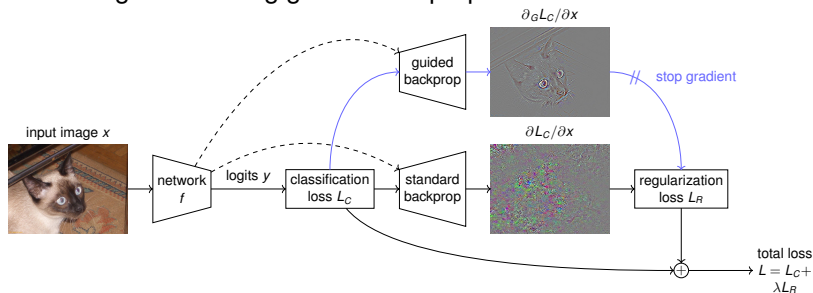
Formalized

Goal: Denoise the gradient during training using guided backprop.
How? Regularize using guided backprop.

Interpretable Gradients

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Interpretable Gradients

Smooth Gradient as Regularization

Assuming target labels T :

$$L_R(X, \theta, T) = \frac{1}{n} \sum_{i=1}^n E(\delta x_i, \delta_G x_i), \quad (11)$$

Interpretable Gradients

Smooth Gradient as Regularization

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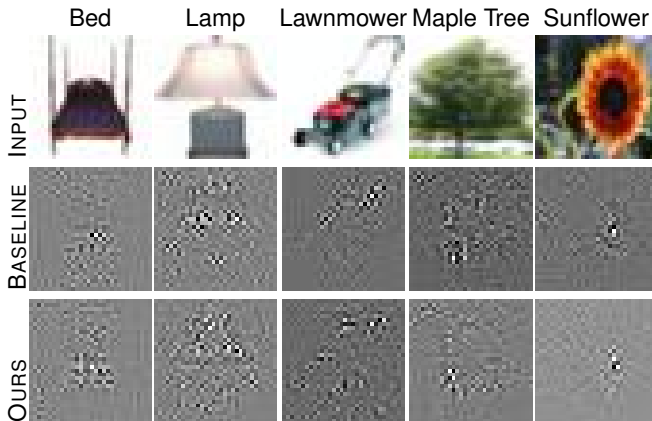
$$L_R(X, \theta, T) = \frac{1}{n} \sum_{i=1}^n E(\delta x_i, \delta_G x_i), \quad (11)$$

And the total loss:

$$L(x, \theta, t) = L_C(x, \theta, t) + \lambda L_R(x, \theta, t), \quad (12)$$

Qualitative Experiments

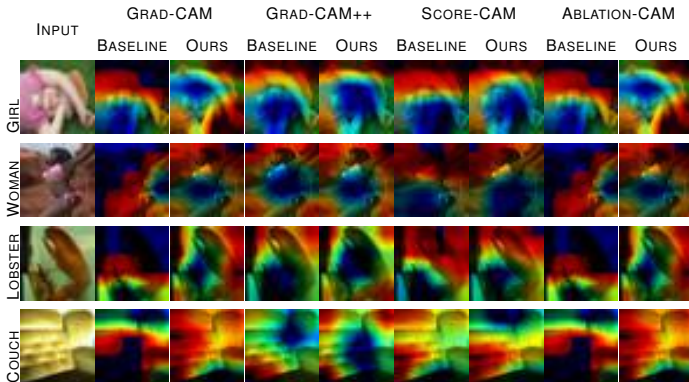
Denoising effect



Results

Qualitative Experiments

CAM Visualizations



Results

Quantitative Experiments

Recognition Metrics

RECOGNITION METRICS			
MODEL	ERROR	λ	Acc $\uparrow$
RESNET-18	-	-	73.42
	COSINE	7.5×10^{-3}	72.86
MOBILENET-V2	-	-	59.43
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Recognition properties are maintained.

Results

Quantitative Experiments

Interpretability metrics

INTERPRETABLE RECOGNITION METRICS							
RESNET-18							
METHOD	L_R	AD↓	AG↑	AI↑	INS↑	DEL↓	
GRAD-CAM	-	30.16	15.23	29.99	58.47	17.47	
	COSINE	28.09	16.19	31.53	58.76	17.57	
GRAD-CAM++	-	31.40	14.17	28.47	58.61	17.05	
	COSINE	29.78	15.07	29.60	58.90	17.22	
SCORE-CAM	-	26.49	18.62	33.84	58.42	18.31	
	COSINE	24.82	19.49	35.51	59.11	18.34	
ABLATION-CAM	-	31.96	14.02	28.33	58.36	17.14	
	COSINE	29.90	15.03	29.61	58.70	17.37	
AXIOM-CAM	-	30.16	15.23	29.98	58.47	17.47	
	COSINE	28.09	16.20	31.53	58.76	17.57	

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Quantitative Experiments

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Interpretable properties are enhanced. Deletion still poses an issue.

Conclusions

- Denoising CNN gradients improve interpretability properties, maintaining recognition.
- Negative gradients contribute to stability during training.
- Resolution matters: small inputs require shallow layer targeting.
- Published in VISAPP 2024.

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Closing Remarks

Conclusion

- Context is important for a prediction. Salient information is found in the background.
- Pooling matters. Small changes lead to enhanced interpretations, with little penalty to predictive power.
- Noisy gradients are a byproduct of training. Negative values act as regularizers.
- Insertion & Deletion are prolematic. OOD data affects the classifier.

Closing Remarks

Perspectives

- Less complex approaches are desired. Enhancing proposed approaches is possible.
- A thorough reality check for *fair* comparison is desired across methods.
- Explanations should extend beyond *groundtruth*. Deployed systems rely on *top predictions*
- Deep models are growing in complexity. Designing approaches to understand them is sought after.
- *For me?*

Closing Remarks

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- Explanations should extend beyond *groundtruth*. Deployed systems rely on *top predictions*
- Deep models are growing in complexity. Designing approaches to understand them is sought after.
- *For me?* I want to pursue a career in research, either in academy or industry.

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